

Akesh Mallia

akeshmallia@gmail.com
470-263-3222
www.linkedin.com/in/akeshmallia

EDUCATION

University of Wisconsin-Madison, Madison, WI

Bachelor of Science in Mechanical Engineering

May 2028

- GPA: 4.00/4.00, Sophomore Research Fellow, Wisconsin International Scholar
- Relevant Coursework: Fluid Dynamics, Thermodynamics, Dynamic Systems

EXPERIENCE

Rocket Lab, Silver Spring, MD

R&D Engineering Intern, Separation Systems Division

June 2026 – Present

- Implementing a hydraulically actuated pull test fixture to measure preload on a separation system
- Designing a PID control system in LabVIEW to automate load application and decrease runtimes
- Modeling in Siemens NX/Teamcenter PDM to streamline procurement and assembly of the fixture

UW-Madison Department of Mechanical Engineering, Madison, WI

Undergraduate Researcher in Dynamic Structures and Data (DSD) Lab

January 2025 – Present

- Studied non-linear vibrations through shaker table analysis of 3D printed stadiums
- Conducted testing by configuring LabVIEW/NI DAQ systems, and processing FFTs using MATLAB
- Performed ANSYS Mechanical harmonic response studies to compare with experimental results

Student Assistant for Intro to Mechanical Engineering (ME 201)

January 2025 – Present

- Instructed weekly labs for 20 students, teaching SOLIDWORKS, statics, and microcontrollers

FTS International, Chantilly, VA

Engineering Intern

June 2025 – August 2025

- Developed a Python-based simulation to assess the impact of satellite signals intelligence (SIGINT)
- Applied satellite dynamics and radio frequency (RF) link budget analysis to generate orbital ground tracks, geolocation uncertainty maps, and signal-to-noise ratio (SNR) diagrams

NASA Goddard Space Flight Center, Greenbelt, MD

Systems Engineering/Project Management Intern

June 2024 – August 2024

- Led 11 interns, through systems engineering and project management, in developing a wireless power transfer CubeSat mission using a collimated LED beam
- Fabricated lens mounts with Onshape modeling and 3D printing, enabling on-ground beam optics test with measurable power transfer between two flat sats

Payload Engineering Intern

June 2023 – August 2023

- Integrated a particulate matter sensor and camera into 3 CubeSats
- Developed data extraction method on a web-based cloud server using Secure Shell connection

ORGANIZATIONS

Wisconsin Space Program, *Propulsion Team and Incoming President*

- Designed, manufactured, and validated a pneumatic quick-disconnect mechanism, successfully ensuring the remote detachment of a 4,000 psi GN2 fill line before launch
- Sized an orifice and integrated regulators, hoses, and fittings on a GSE cart, enabling smoother cold flows and launch operations
- Launched Northstar, the first liquid rocket in Wisconsin state history, at FAR in May 2026

Societies: Pi Tau Sigma (2025-Present), Tau Beta Pi (2025-Present), Japanese Student Association

SKILLS

CAD (NX, SOLIDWORKS, Onshape) • FEA (Ansys Mechanical, SOLIDWORKS Simulation) • Manufacturing (Mill, Lathe, MIG Welding) • Programming (Python, MATLAB, EES) • LabVIEW • NI Hardware